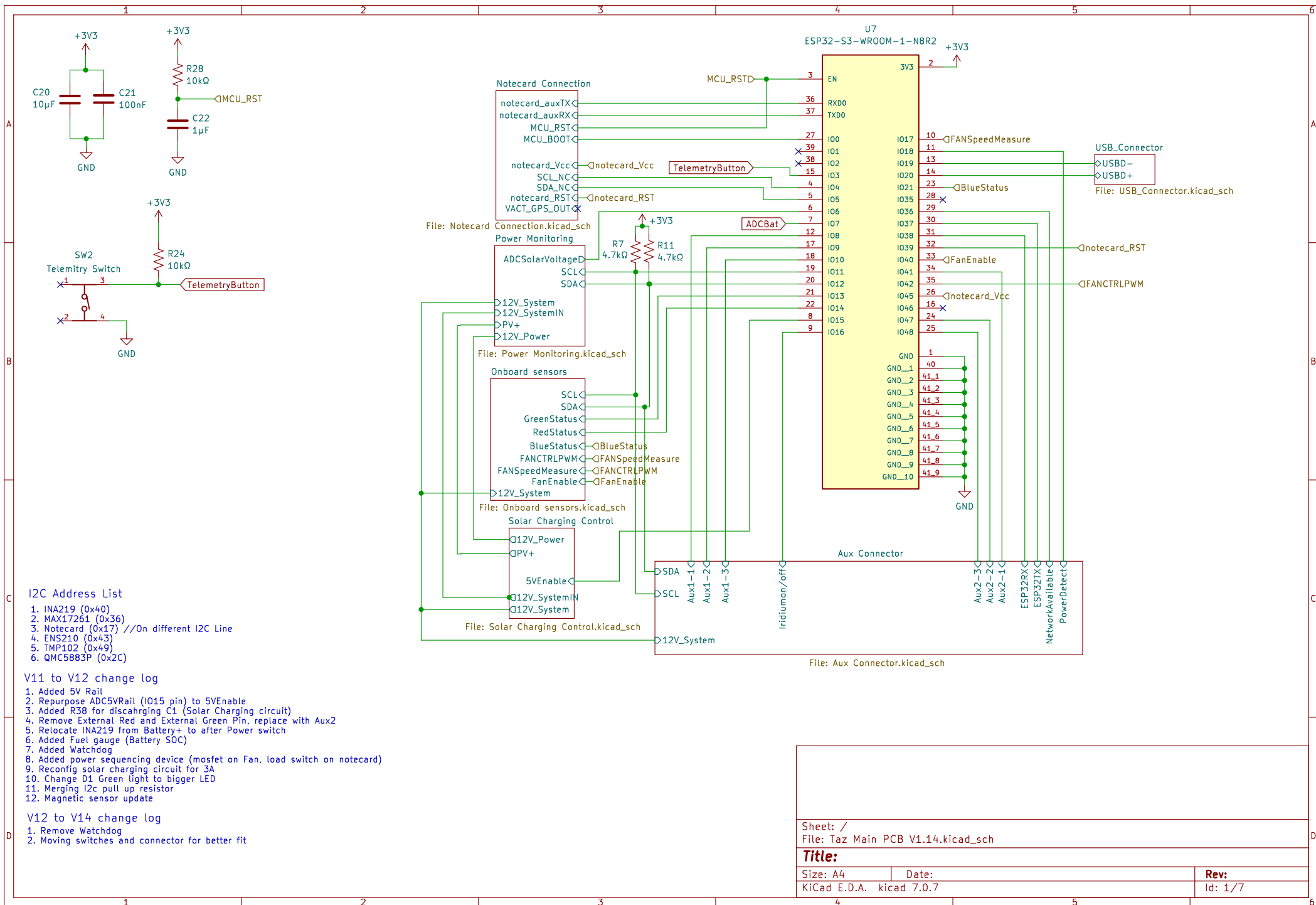




Using model from
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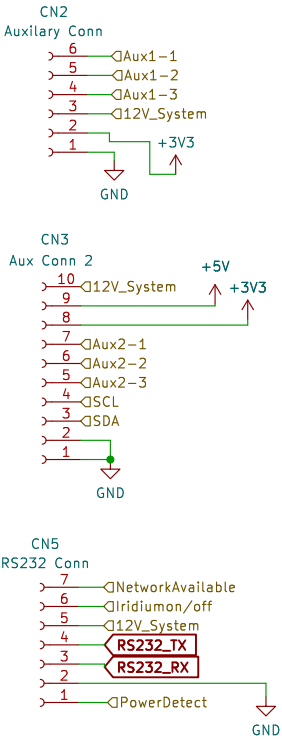
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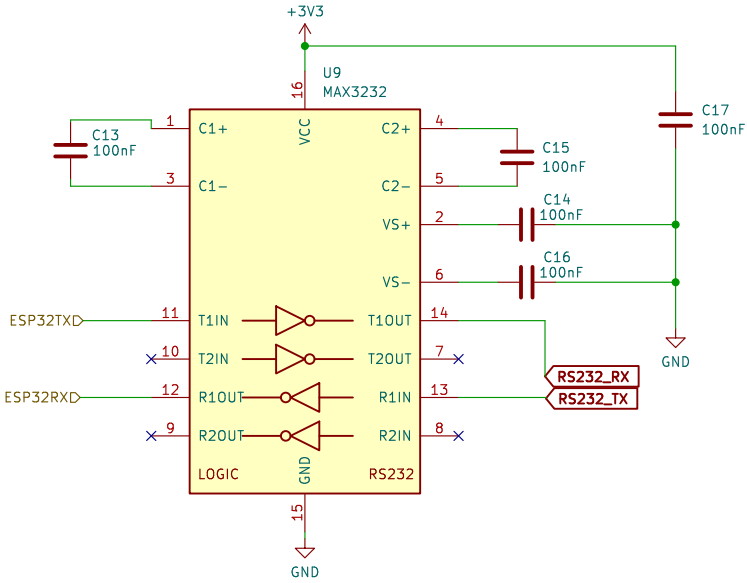
Id: 2/7

Aux connector

- Current option:
1. CN2, AUX1 (Stepppter motor connector)
2. CN3, AUX2
3. CN5, Satalite module connector (RS232)



RS232 Driver and Receiver



Sheet: /Aux Connector/
File: Aux Connector.kicad_sch

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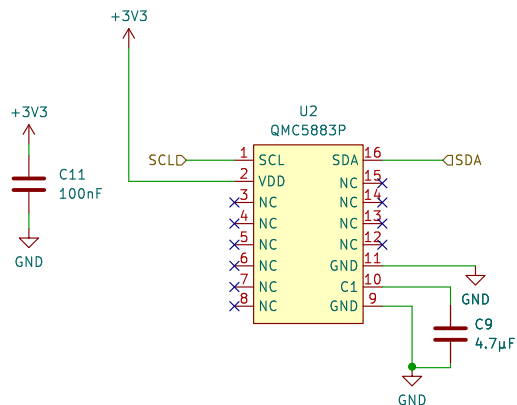
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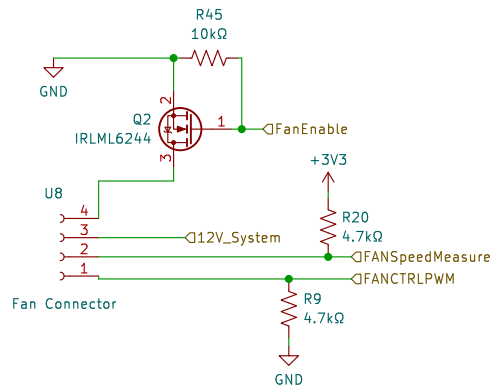
Onboard sensors

Current sensor on the board:
1. Compass/accelerometer
2. Temperature and Humidity sensor
3. LED Status
4. Fan

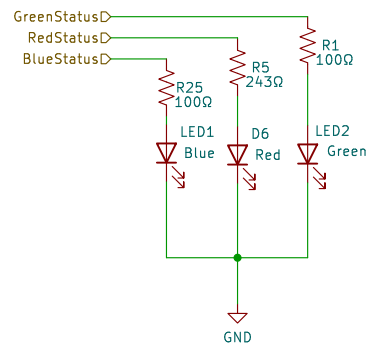
Compass



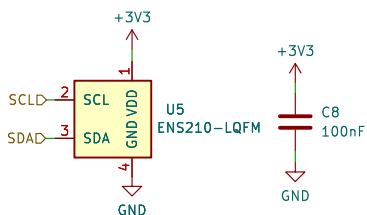
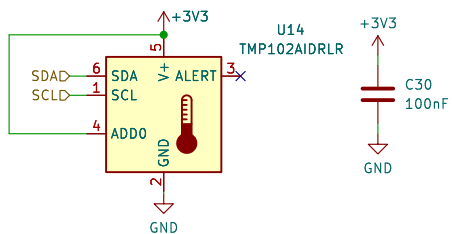
Fan Connect



Status LED



Temp and Humidity Sensor



Sheet: /Onboard sensors/
File: Onboard sensors.kicad_sch

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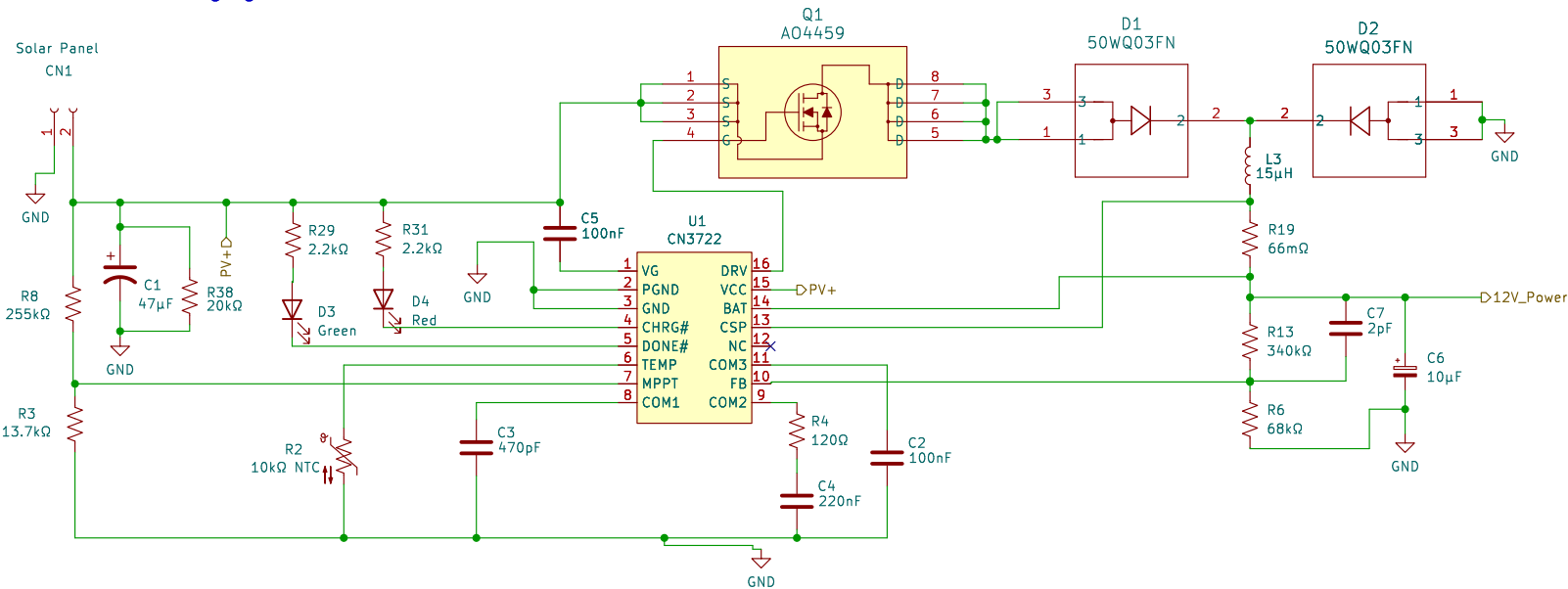
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Id: 4/7

Solar Power Charging



L3 Table
PCB v11 22μH Rcs 100mΩ (Designed for 2A)

Charge Current Calculation
I_{ch} = 200mV/R_{cs}

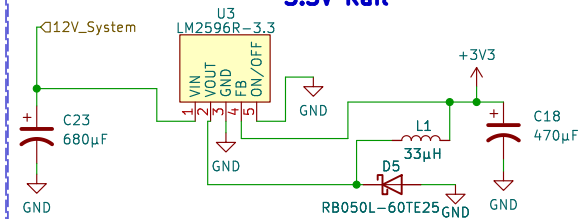
Inductor selection guide		
Charge Current	Input V	Inductor Value
1A	>20V	40μH
	<20V	30μH
2A	>20V	20μH
3A	<20V	15μH
4A	<20V	10μH

Power Switch

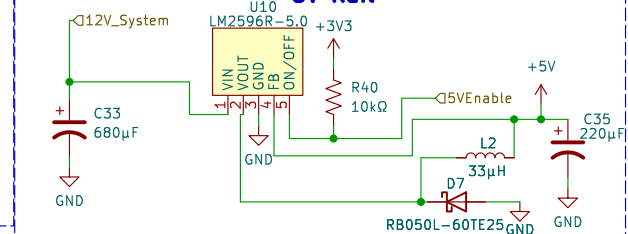
12V_Power = Output from Battery
12V_System = Input to Sensor



3.3V Rail



5V Rail



To turn LM2596R on, the ON/OFF pin should pull LOW

Sheet: /Solar Charging Control/
File: Solar Charging Control.kicad_sch

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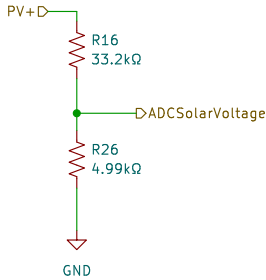
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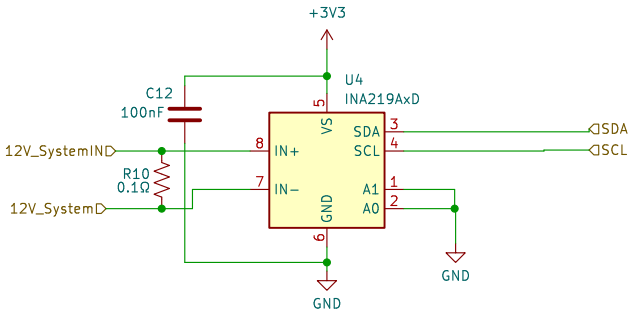
Power Monitoring

- Current Measuring:
1. Solar Panel Voltage (through ADC pin)
2. Battery Voltage and Current after power switch(INA219)
3. Fuel Gauge on the low side of battery

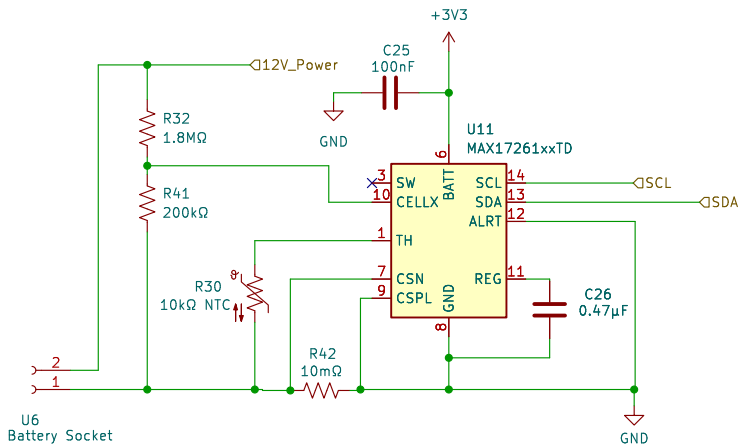
Measuring Solar panel Voltage



Measuring Voltage and Current After Power Switch



Battery Fuel Gauge



Sheet: /Power Monitoring/
File: Power Monitoring.kicad_sch

Title:

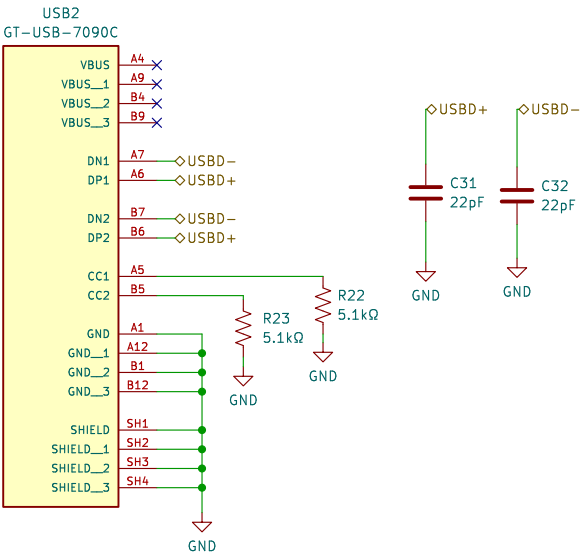
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Id: 6/7

USB Connector

Current connection allows JTAG connection throguh USB



Sheet: /USB_Connector/
File: USB_Connector.kicad_sch

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KiCad E.D.A. kicad 7.0.7

Date:

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